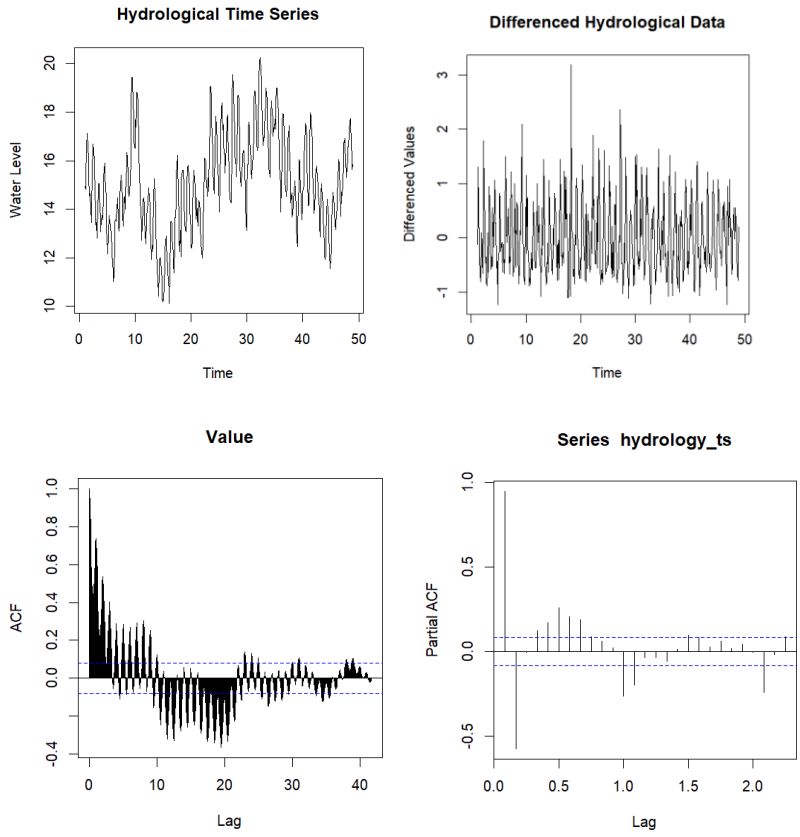
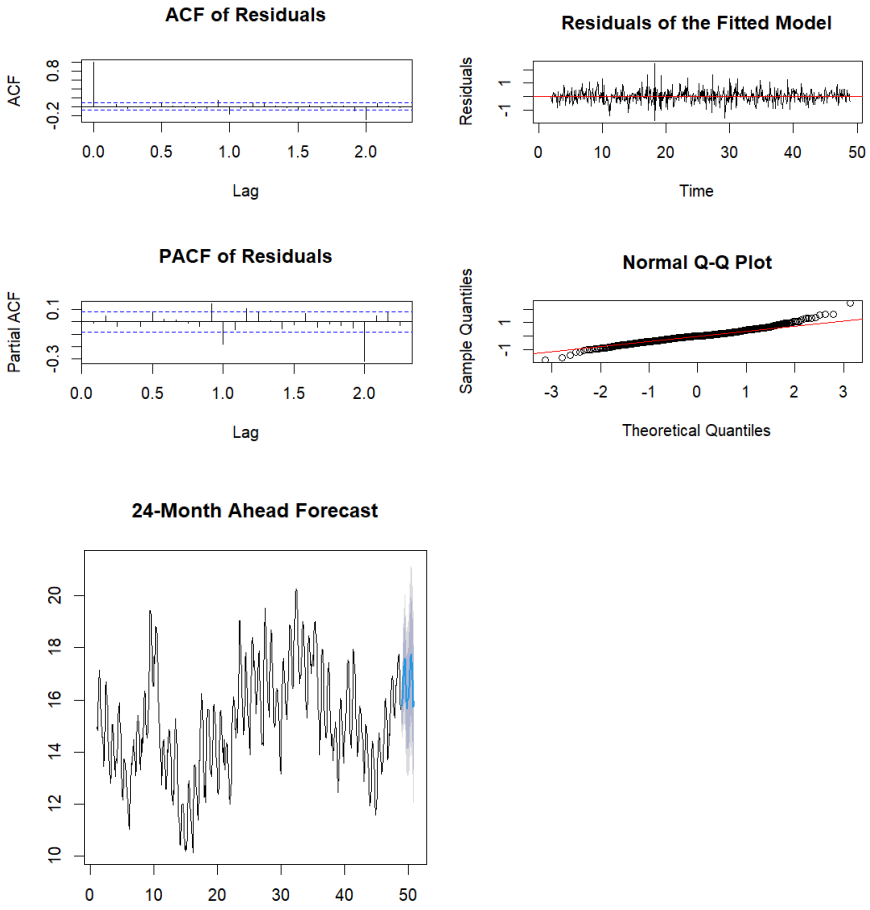
Scenario 1: Hydrological Forecast

- **Model Description**: The ACF plot is particularly useful in identifying seasonality. PACF plot show a clear spike at lag = 2, which suggests an AR(2) term. Hydrology is a monthly data and we see spikes at lag 12 for monthly data in ACF plot. By the help function auto.arima(seasonal = TRUE), we choose Auto choice model: SARIMA(2,0,0)(1,1,0)[12].
- **Forecasting Process**: We simply use function forecast to generate the 24-month ahead forecasts and calculated the prediction intervals.
- **Diagnostic Plots**: Include plots of the time series data with the forecasts and 95% confidence intervals. Discuss model diagnostics, such as ACF/PACF plots, residual analysis, and any checks for stationarity. After modeling, residual of ACF, PACF help verify that the residuals have no remaining autocorrelation. Residual check ensures that the model's errors are random, uncorrelated, and normally distributed, which supports the validity of the model. Box-Ljung test p-value far higher than 0.05 (0.47). This suggests that the residuals do not show significant autocorrelation, meaning that our model is well-specified.

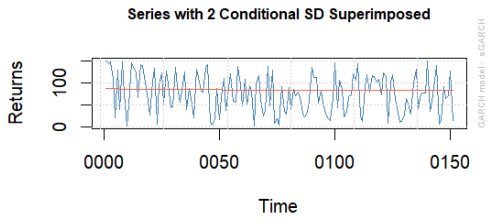


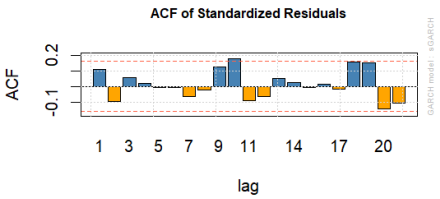


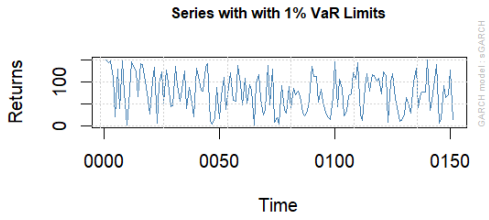
Scenario 2: Financial Risk Forecast

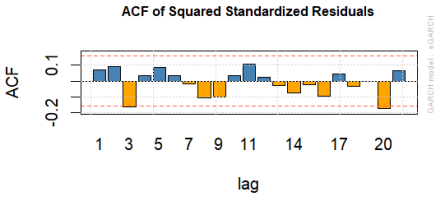
- **Model Description**: Financial time series, such as stock returns, typically exhibit volatility clustering, where periods of high volatility (large fluctuations) tend to cluster together, followed by periods of low volatility. This means that the variance of the series is not constant over time but changes in a predictable way. The GARCH model is specifically designed to model and forecast such changing volatility. It allows the conditional variance (i.e., the volatility) to depend on past values, capturing the volatility clustering observed in financial data.
- **Data preparation:** Forecasts for stock 1-40 should be stored in the columns of a 10x40 dimensional matrix. Set the directory where my stock files are located. List all .txt files that start with "stock" in the directory. Files are disordered and we utilize “mixedsort” function. Now it comes with right order. Loop over each stock file to fit the model and generate forecasts.

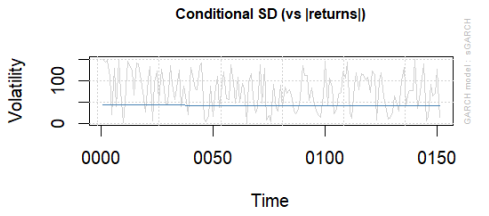
- **Forecasting Process**: Detail how the 10-step ahead quantile forecasts were produced for the stocks.
- **Analysis**: After modeling, residual of ACF, PACF help verify that the residuals have no remaining autocorrelation. Residual check ensures that the model's errors are random, uncorrelated, and normally distributed, which supports the validity of the model. Sample quantile show close normal distribution.

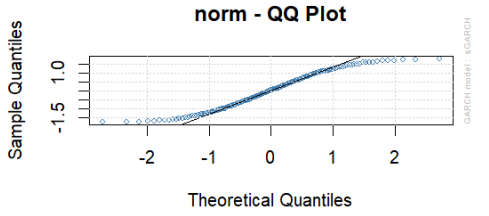


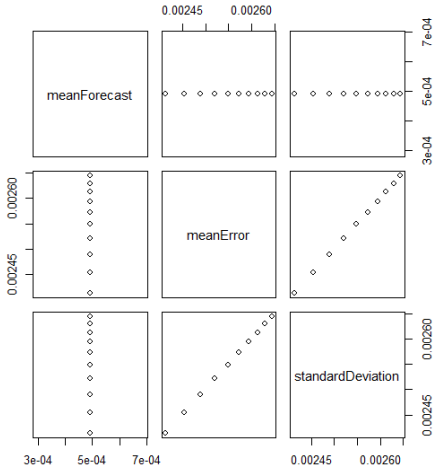










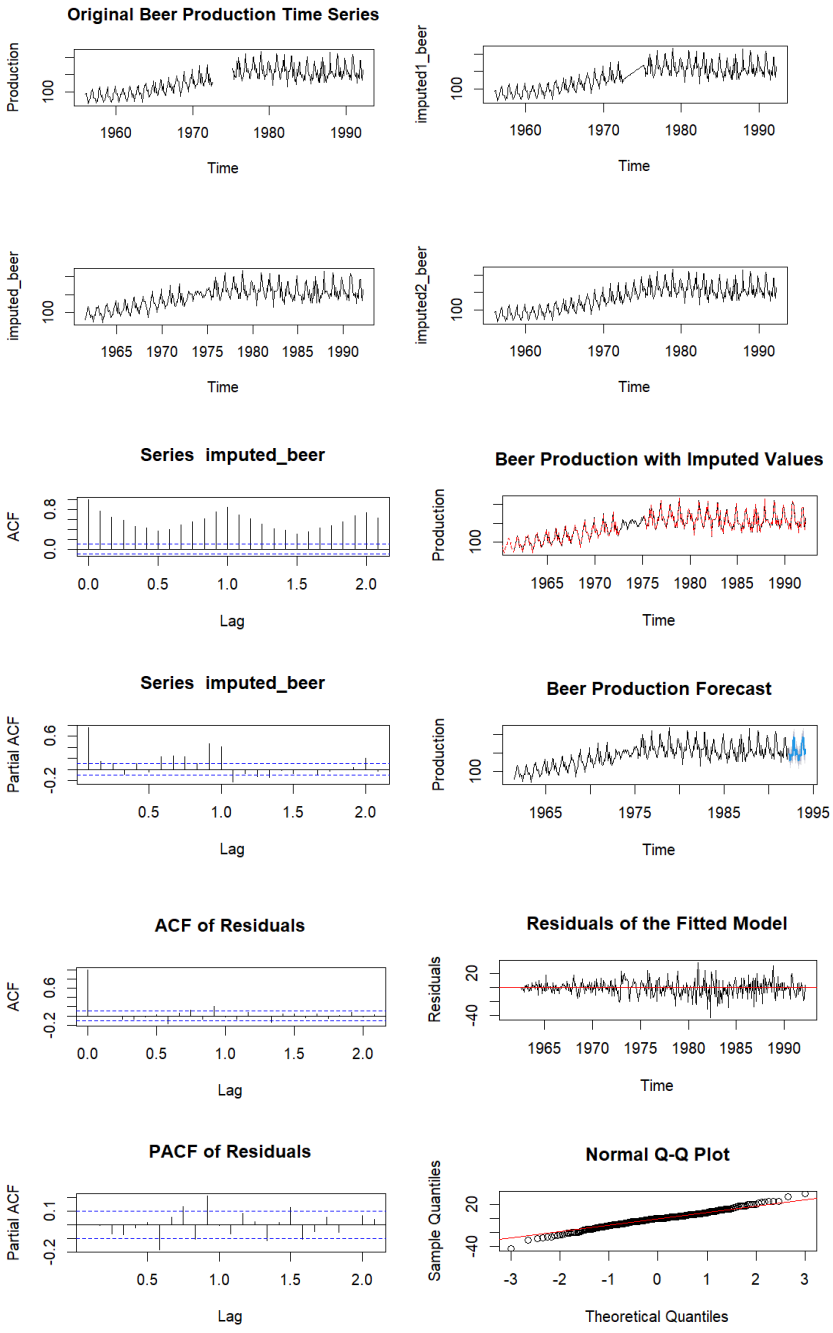


Scenario 3&4: Imputation and Multivariate Time Series Forecasting

- **Data preparation:** Since they have different start date. Misaligned time series can lead to incorrect analyses, especially when performing imputation

and forecasting that rely on the temporal relationships between the series. Determine the common start date (latest start date among all series). We align all the data start with the latest beginning date 1961-07.

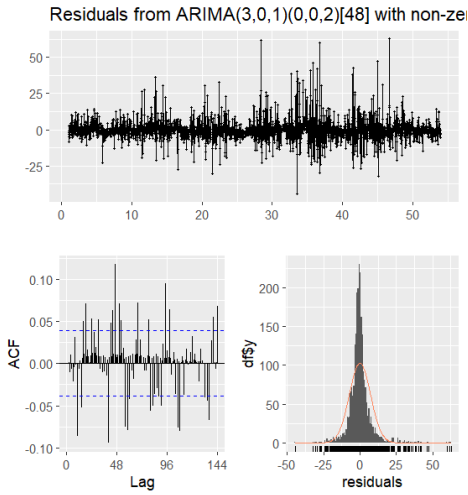
- **Imputation Process:** We choose missForest. Since data has complex, non-linear relationships that are not well captured by linear models. I want a method that is robust and generally performs well across a wide range of data types and missing data patterns. We also use functions na.approx and na.interp to compute beer imputation1 and imputation2, which doesn't consider other data. We plot these two to make comparison. And original imputation show a similar trend, which looks good.
- **Model Description:** The ACF plot is particularly useful in identifying seasonality. PACF plot show a clear spike at lag = 1. And we know that all the data are monthly generated. By the help function auto.arima(seasonal = TRUE), we choose Auto choice model: SARIMA(2,1,2)(0,1,2)[12]
- **Forecasting Process**: We simply use function forecast to generate the 24-month ahead forecasts and calculated the prediction intervals.
- **Diagnostic Plots**: Include plots of the time series data with the forecasts and 95% confidence intervals. Discuss model diagnostics, such as ACF/PACF plots, residual analysis, and any checks for stationarity. After modeling, residual of ACF, PACF help verify that the residuals have no remaining autocorrelation. Residual check ensures that the model's errors are random, uncorrelated, and normally distributed, which supports the validity of the model. Box-Ljung test p-value = 0.99, which shows highly uncorrelated.

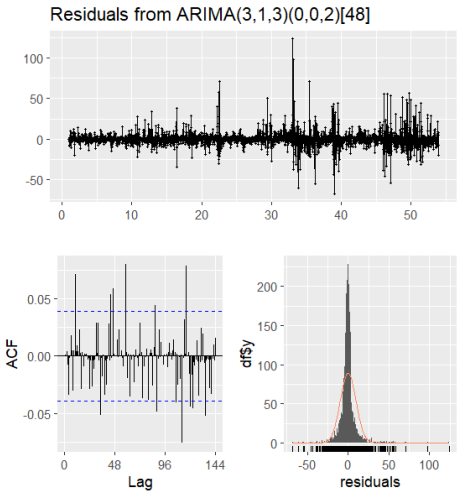


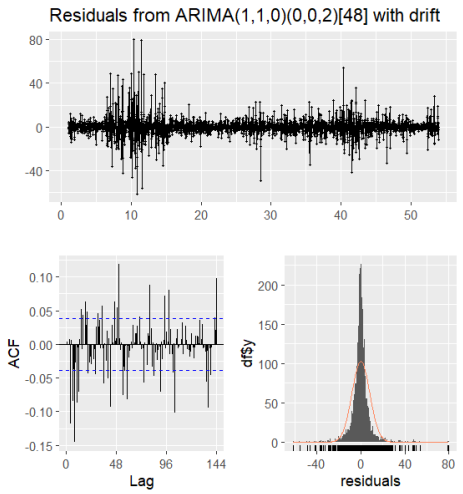
5. Scenario 4: Long Horizon Pollution Forecasting

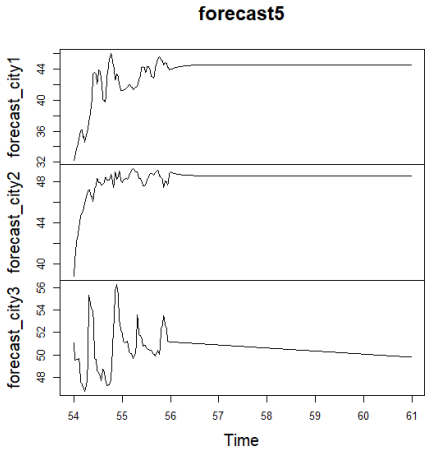
- **Data Preparation:** Forecasts for cities 1-3 should be stored in the columns of a 336x3 dimensional matrix. By setting the frequency correctly, any seasonal effects or daily patterns in the data are accurately modeled and forecasted. frequency = 48: Appropriate for half-hourly data because it reflects the number of half-hour intervals in one day.

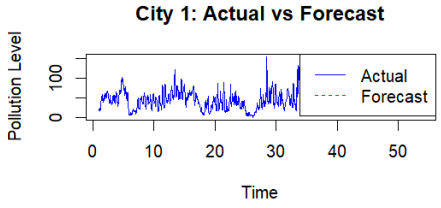
- **Methodology**: Since the data is extremely huge. I choose to write a function to generalize the whole modeling process. By the help function auto.arima(seasonal = TRUE), we choose Auto choice model.
- **Forecasting Process**: Detail the 336 half-hourly steps and generate it into a matrix.
- **Analysis**: The residual plots show that the errors are generally centered around zero, with no obvious patterns, indicating that the model has captured the main structure of the data. The ACF plots of the residuals indicate no too significant autocorrelation remaining, which suggests that the model is appropriately specified. But it seems that auto model may not be the best choice. But because data has too many steps and a larger scale. I think it is acceptable to improve the code and analysis efficiency. The plot now includes the actual vs. forecasted values for City 1 as an example, which can be more informative when assessing the model's performance. It follows it previous trend.

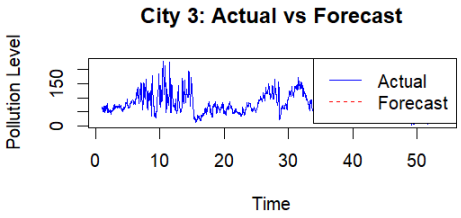

Residuals from ARIMA(3,0,1)(0,0,2)[48] with non-ze
ACF
Lag
df$y
residuals

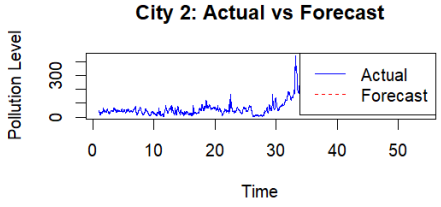

Residuals from ARIMA(3,1,3)(0,0,2)[48]
ACF
Lag
df$y
residuals


Residuals from ARIMA(1,1,0)(0,0,2)[48] with drift
ACF
Lag
df$y
residuals


forecast5
forecast_city1
forecast_city2
forecast_city3
Time


City 1: Actual vs Forecast
Pollution Level
Actual
Forecast
Time


City 3: Actual vs Forecast
Pollution Level
Actual
Forecast
Time


City 2: Actual vs Forecast
Pollution Level
Actual
Forecast
Time